Implementation of Electronic Records Management Systems: Potential and Challenges
A Case Study of the Water and Power Development Authority (WAPDA) in Pakistan

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Abstract. The state of record-keeping in the majority of local government organizations in Pakistan remain outdated where conventional systems are still followed for daily processes due to the vast amount of legacy systems and red-tapism. This research conducts a case study to understand the current state of record-keeping in a local government organization and the challenges in implementing an electronic records management system (ERMS); to promote the concept of good governance with the aid of transparency and accountability. A government department in Pakistan ,i.e., the Water and Power Development Authority (WAPDA), is the selected case study; qualitative data were collected from expert interviews and employee questionnaires, which have been analyzed with tool-based thematic analysis. The proposed state of processes is displayed through a To-Be model by a visual representation of the advantages that an ERMS system can provide for a proper flow of steps and timely delivery of services.

Keywords: Electronic Record Management System, ERMS, Information and Communications Technology, ICT, information technology, good governance

1 Introduction

In the digital world of today, the adaption of an electronic records management system (ERMS) is no longer a matter of choice, but a necessity. This is because of the radical penetration of computers into offices of public and state departments. IT systems based on computers have revolutionized the way of carrying out day-to-day tasks. This has resulted in the proliferation of data files in the system, and in turn, made it difficult to appraise and preserve these files [1]. Records maintained in a government organization are often the most crucial source for the processes of accountability. In a government organization such as the Water and Power Development Authority (WAPDA) of a developing country such as Pakistan, ERMSs have not been integrated nor been considered as part

of the strategic management program. Therefore, this research analyses the current state of record-keeping in WAPDA, all the possible challenges that may be faced in its process of ERMS adaption and the potential for increasing efficiency in the daily workflow. A case study approach has been utilized along with interviews from officials and questionnaires from employees. This paper is part of the Master Thesis which was submitted and defended in 2019 in Tallinn University of Technology [2]. The outcome of this research can serve as practical knowledge for WAPDA with recommendations and re-designed 'new connection request' as input for further improvement of the other processes in the organization.

The rest of the paper is organized as follows, in Sect. 2 related work as well as case is briefly described. Section 3 provides the details of the methodology used for this research. Section 4 shows the results from the interviews and questionnaires, which are discussed along with recommendations in Sect. 5. And lastly, Sect. 6 presents the conclusion.

2 Related Work

2.1 Good Governance and ERMS

The idea behind the research paper is to facilitate the ideology of "Good Governance" in developing countries, through the application of necessary digital systems in government organizations. A modern form of government is about the efficiency of the system and accountability between the citizens and the government. According to Grindle, "the principle of accountability in the real sense, emphasizes answerability for the use of state resources and assets earmarked for specific purposes, subject to the laws and their requirements" [3]. An ERMS helps to build interoperable systems that leads to accountability [4]. Moreover, conformity to rules and regulations can be ensured through transparency as Nanda states, "in fact, the processes of decision-making, the ultimate decision reached and the government actions taken are expected to be made open and subject to check by other organs of government and non-governmental organizations" [5]. In the decision-making process, transparency is the free flow of information [6] that is easily accessible and understandable to the audience that may be affected by it. According to Nanda "effectiveness is primarily the process of doing the right things, while efficiency is primarily the process of doing things right" [5]. A proper action plan should be developed by the government that takes care of the necessities of the societies while making sure that the resources are put for use in an efficient manner.

2.2 ERMS in Pakistan

Record Management is defined as "the planning, controlling, directing, organizing, training, promoting, and other managerial activities involved in records creation, maintenance and use, and disposition in order to achieve adequate and proper documentation of the policies and transactions ... and effective and

economical management of its operations." [7]. The basis of any public sector organization is records management; this is because records are meant to be the "institutional memory"[8]. It is through effective implementation of an ERMS that the efficiency of a government in providing services to its citizens is increased [9].

The introduction of an ERMS in any organization brings a radical change in the organizational culture, the responsibilities, polices, and the standard procedures. The capability that an organization depicts in managing this change is the determining factor of the process of implementation [10]. According to the study conducted by Zinner and Andersen, the system that was followed throughout the Punjab Government departments was the one developed almost 80 years ago by the British [11]. With the growth in the government operations and the rapid increase in the responsibilities assigned to each department, paper records caused deterioration of the efficiency level as they were stored in big filing rooms with no system to track them and required immense manual work in case if a document had to be retrieved. There was no formal nor proper procedure for the record-keeping in the rooms, in the desks, or the offices where the records were stored [11]. The study further explains that the major constraint was the lack of technical abilities in the IT Department of Punjab. There is near to none experience for such initiatives in the country. The staff is trained to follow manual procedures and shifting to the electronic system means providing them with the technical skills and training them accordingly [11].

The implementation of ERMS has remained limited in Pakistan, even though, it has increased the level of efficiency and effectiveness in a few government departments such as NADRA [12]. According to Yahaya, the workstations in most of the government departments in Pakistan generate a huge volume of documents and emails [13]. The issue here is to be able to manage these volumes of electronic documents and records. Moreover, to implement an ERMS, some important factors need to be analysed so that the records are obtained, created and also managed as per the policies of the department. The factors that may pose as challenges to the ERMS implementation are presented in Table 1.

2.3 Case Background

WAPDA was created in the 1958 and has the responsibility to coordinate and give out unified directions for the development of plans related to the Power and the Water Sector. In 2007 the organization was split where PEPCO (Pakistan Electric Power Company) was made responsible for the production of Power, its supply, transmission and billing whereas, WAPDA was given the responsibility of water and hydropower development. The main objective of the organization is to make sure that the supply of electricity is reliable throughout the country. As per the statistics of 2012, WAPDA has 20.12 million users with a major share of 43% in the domestic sector, 26% in the industrial sector and approximately 12% in the agriculture sector [21]. WAPDA has monopolized the distribution and the transmission networks. When WAPDA was unbundled, there were nine distribution companies (DISCOs) created, four of the generation companies (GENCOs),

IT literacy	No formal IT curriculum in educational institutes. Lack of training and skillset further translated into the workforce of an organization.
E-Readiness	Tools that aid in assessing the e-readiness that can be divided into two main categories: one that assess the basic infrastructure of the country and the second that assesses the ability that a nation possesses in order to take advantage from the ICTs.
IT infrastructure	An IT infrastructure means the technical bases like hardware, the communication technology, software, basic applications and the capability of the user to use these applications [14].
Co-operative Management	If there is no support from the management, it may end in preventing the users to use the particular system [15]. With the implementation of IT, the power enjoyed by the top management is challenged, which is why many times they are reluctant to digitalize the system [16].
Training	Training can be done by introducing in-service programs, workshops, conferences etc. that cover the basic principles and functionalities of an ERMS [17]. These trainings can be time consuming and expensive for the workforce that is not technologically advanced.
IT Policy	It is imperative to have policies in place for organizations to take digitalization seriously [18]. Different countries has different policies to ensure the safety of data as well as to prove the authenticity and reliability of the records [19].
Culture	Hofstede defines culture as "patterned ways of thinking. . . the essential core of culture consists of traditional (i.e. historically derived and selected) ideas and especially their attached values." [20] In Pakistan, paper is considered as a trusted form of documentation against electronic records.

Table 1: Challenges in ERMS implementations.

and a transmission company. After the transmission, power is then handed over to the eight distribution companies (shown in Fig. 1) in order to make sure that it is evenly distributed amongst the consumers of the state. For the purpose of the study, the services provided by Gujranwala Electric Supply Company (GEPCO) will be discussed for a thorough understanding of the operations in the department; translating to WAPDA eventually.

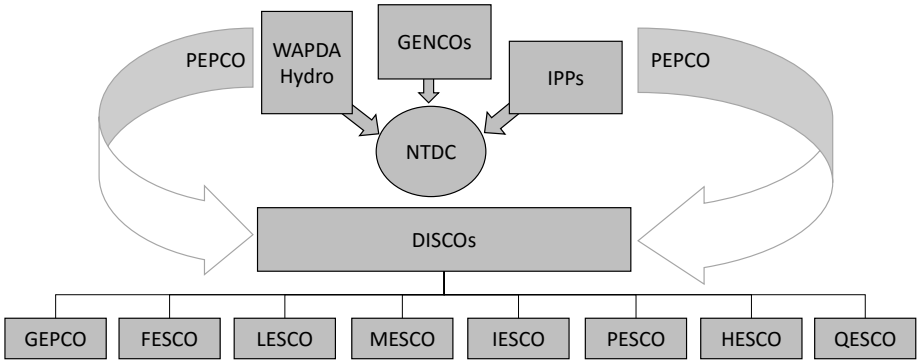

Fig. 1: Divisions and subdivisions of WAPDA.

3 Research Methodology

This research utilizes the case study approach in order to understand the current state of record keeping in the country and the challenges in implementing an ERMS. The case study for this paper is WAPDA, a government department in Pakistan. Data in this case was collected from employees at different levels of the department in order to determine the current situation of record keeping and services provided. Interviews were conducted from the following officials at GEPCO:

- Sub-divisional Officer (SDO)
- Executive Engineer (ExEn)
- Customer Service Specialist (CSS)

Interviews (approx 40min long) were then analyzed using an open source qualitative data analysis software, RQDA to carry out a thematic analysis. On the other hand, a questionnaire was also used in order to gather information from the employees of WAPDA and its subdivision GEPCO. A total of 50 employee responses were recorded.

4 Results

4.1 Interview Results

Different codes that were generated during the analysis using RQDA are shown in Fig. 2. These codes were then used to generate themes which are shown in Table 2.

4.2 Questionnaire Results

All of the employees agreed to the idea that technology in this era has become an essential part of our daily routine. The responses to the main issues that currently WAPDA faces in its daily work-flow were mixed where most of the respondents agreed that the lack of material resources (70%) and skilled labour (62%) was to blame. 72% of the respondents claimed that poor customer support was the reason for the WAPDA's reputation to deteriorate. The majority of employees believed that the method of record keeping can be improved and that ERMS can serve as an effective solution in order to increase the efficiency. While most of the respondents claimed that through the implementation of ERMS there will be transparency of processes and is essential to increase efficiency, many believed that it is an expensive process and that unskilled labour might lose their jobs in the organization. On the required IT infrastructure, 52% respondents believed that the necessary infrastructure is not present. 54% of the respondents agreed that WAPDA employees do not have any sort of IT skills which may become a hindrance to the ERMS implementation and therefore, 68% agreed that there will be resistance to this change.

Processes and Services at GEPCO	– New connections request: • Office visit • Website Even after a new connection request is filled online, the rest of the process remains manual where most of the communication between the team is through WhatsApp messages or telephone calls. – Complaint and Customer Services: • Online portal • Manual entry in register These complaints travel from the clerical staff to the SDO and then to the ExEn, where documents are exchanged through postal service. If the complaint is not redressed, customer has to pay multiple visits to the office and every time his complain is fetched from a register that can be time-consuming. – Billing Services: Lineman or staff take meter readings through their phones, which are then communicated through WhatsApp messages and then updated into the systems manually in order to generate manual as well as online bills.
Barriers to ERMS implementation	– Shortage of labour which also shows that the processes are redundant which means a large workforce is required (SDO) – Difficult to process the new connection requests, or cater to the complaints of the individuals when the necessary equipment is not readily provided by the state. (SDO) – Most of the workers have been working the same job since more than three decades where no training is provided to them in order to improve their skill set as per the current requirements. (ExEn) – Power is distributed through hierarchy that often leads to misuse of power and authority. Bribery, corruption and nepotism. (SDO) – Citizens of the state unaware of their civic responsibility where even the online portals are exploited in order to register false complaints and create issues for a department which already is short of workforce and resources. (CSS) – Internet was only available to the managerial level of the department that was used seldom to send emails. ID documents shared through postal service. (ExEn) – No archival repository besides a tape drive and registers. (SDO) – The clerical staff is barely equipped to use a computer, most of the details and complaints are still jotted down in a register. (CSS)
Good governance: Transparency, accountability and efficiency	– Follow up on a complaint requires periodic visits to the office or phone calls. Staff already overworked and do not cater to each inquiry; are often found rude while providing inadequate responses. This has slowed down the processes, created undue frustration, encourages bribery culture and resentment of the common public towards the government departments in general and WAPDA or GEPCO in particular. (CSS) – Hybrid system (paper + online) is being used for record keeping. GEPCO uses tape drives in order to save all the data including the sensitive information. There is no cloud-based system or a database that saves the data collected from the public. Tape drives are not reliable and there can be a huge loss of data due to any natural calamity, technical faults or personal favors. (SDO) – Document search is time-consuming, "once we took 5 hours to search a document. It was some bureaucrat's complain so we had to." (CSS)

Table 2: Themes

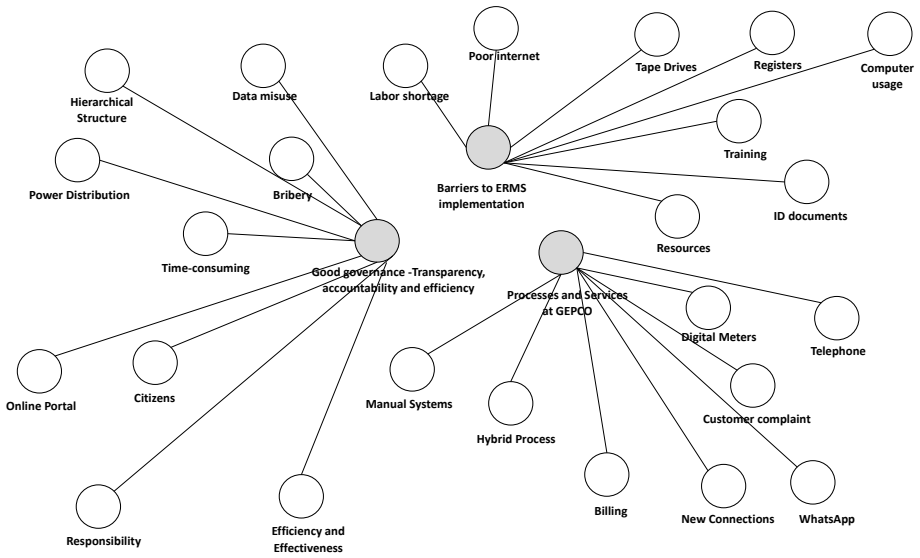

Fig. 2: Thematic map.

5 Discussion and Recommendations

The research suggested that to effectively enroll an ERMS in a government department such as WAPDA, it is of paramount importance to first analyze the existing hybrid system for records management. For an effective ERMS implementation, GEPCO needs to update its existing computer infrastructure, introducing systems that are up-to-date and networks that will be able to support the ERMS. Different servers such as software, processing power etc., has to be determined that could be used for an ERMS implementation according to the department requirements and server storage capacity has to be defined. According to the interviews, internet connection was poor and barely used to exchange emails. The documents exchanged over emails also have to be analyzed to distinguish which emails are considered as records and then a records management policy has to be devised to retain email records. The highest bandwidth of network servers as well as multifunctional devices have to be put in place to accommodate high volumes of records.

Furthermore, the management needs to be trained beforehand, to make sure that they are capable enough to handle the change that an ERMS system will bring to the organization. If workers are included in the ERMS implementation process and planning, it is more likely that they are willing to get training as well as accept change. Moreover, with the ERMS implementation, not only the work procedures will change but also the organizational culture. If an ERMS is used, it can eventually lead to accountability and transparency which means power will be redistributed and may shift down to the system handlers and users. It

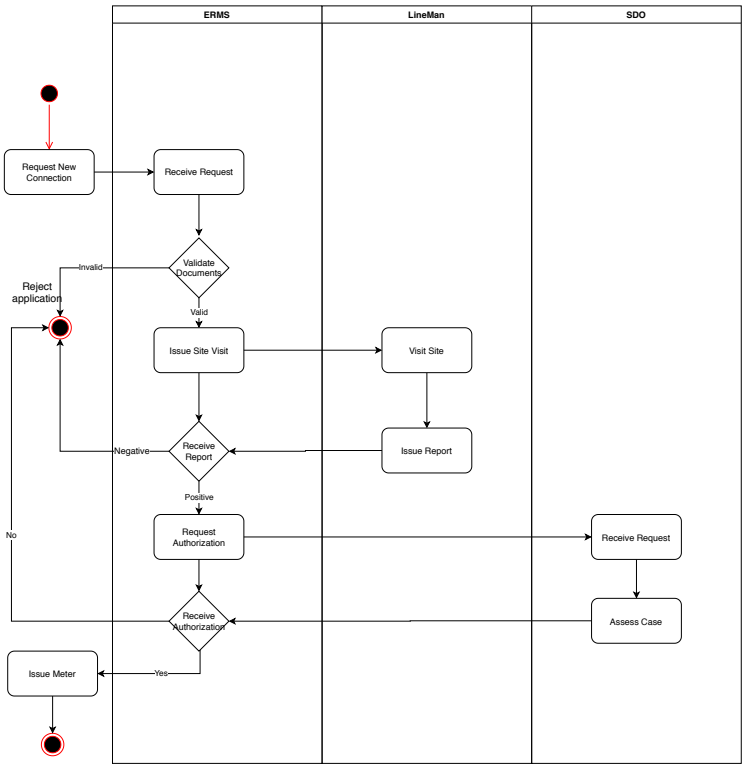

Fig. 3: To-Be Model for New Connection Request at GEPCO.

will have an immense effect on the decision making process when the system takes control to assess the workload and workflow [11]. Pakistan has a lot of potential and many trained professionals in the IT department who understand the need of implementing the ERMS in various government departments. The need here is to make sure that they have enough resources and finances in order to pursue their ideology and introduce the system of digitization in at least the major departments run by the government such as WAPDA. Pakistan can take inspiration from Estonia as "Best Practice" title was won for its transition to a paperless management system in 2009 [22]. The Government of Estonia has always stayed a step ahead and provided the system with a "defined regulations on standard, and pros, procedures, methods, and products to assist the modern IT-evolution processes" [22]. Moreover, an X-road has been established and interoperability has been created to strengthen private public relationship [23] and eID is being used which further compliments digital signatures [24], e-invoicing [25], e-voting etc. Therefore, there are several factors that need consideration in order to make sure that the implementation of ERMS in a government organization is successful which includes:

- Increasing the IT literacy within the organization.
- Determining the e-readiness.
- Developing a reliable IT infrastructure.
- Having a co-operative management.
- Developing training programs for employees to equip them with required IT skills.
- Formalizing IT policies to ensure that the implemented ERMS is utilized as intended.

Each one of these factors has to be systematically researched further to understand the needs and expectations of different stakeholders involved in the ERMS implementation in WAPDA and other government organizations in general.

The purpose here is to design a service that is based on the experiences from the customer-end perspective. By engaging all of the stakeholders in the process of service design, it results in a system that is cohesive and aligns well with the needs of the customers providing them with the better service. Therefore, the To-Be Model, depicted in Fig. 3, has been designed in order to help visualize that implementation of an ERMS that can remove the reasons behind delayed services and the iterations that are redundant. The process for new connection request at GEPCO has been re-designed to shorten the workflow and subsequently to increase efficiency.

6 Conclusion

The purpose of the paper was to analyze the enrolment of an ERMS in a government department in order to ensure efficient processes and services which facilitate good governance, while also analyzing the possible challenges that might arise. It was found out that the implementation of an ERMS and a better

management of public information is capable of increasing transparency and accountability. The lack of IT infrastructure and limited resources is identified as the root cause of outdated record management in WAPDA; therefore, the need for reforms and implementation of an ERMS cannot be overemphasized. The ERMS implementation can be tedious, expensive, time-consuming and cause resistance to change; but it is important to realize that in order to survive in this technologically advanced world, an ERMS is an essential first step towards digitalization.

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